

CHAPTER 24

WATER AND LAND RESOURCES UTILIZATION AND PLANNING

Importance of Water Resources

- India:
 - **2.45%** of world's land resources
 - **4%** of world's freshwater resources
 - **16%** of world population
- Water resources are classified into:
 - **Surface water resources**
 - **Groundwater resources**
- Both are **interdependent**

River Systems and Inland Water Resources in India

- **12 major rivers** with total catchment area of **252.8 million ha**
- **Ganga–Brahmaputra–Meghna system**
 - Catchment $\approx$ **110 million ha**
 - **43%** of total major river catchment
- Other major rivers:
 - Indus – 32.1 million ha
 - Godavari – 31.3 million ha
 - Krishna – 25.9 million ha
 - Mahanadi – 14.2 million ha
- Medium rivers total catchment $\approx$ **25 million ha**
- Largest medium river: **Subernarekha (1.9 million ha)**

Other Inland Water Bodies: Canals, reservoirs, tanks, ponds

- Area under inland water bodies (excluding rivers & canals): **$\sim$ 7 million ha**
- Tanks & ponds:
 - Maximum area: **2.9 million ha**

- Reservoirs:
 - Area: **2.1 million ha**
- Five states (Odisha, Andhra Pradesh, Gujarat, Karnataka, West Bengal) account for **>50%** inland water bodies

Rainfall in India

- Rainfall from:
 - South-West monsoon
 - North-East monsoon
- Average annual rainfall: **~1170 mm**
- Range:
 - Rajasthan (west): ~100 mm
 - Meghalaya: >10,000 mm
- Total precipitation (rainfall + snow + glacier melt): **~4000 BCM**
- Average annual water availability: **1869 BCM**
- Utilizable water resources: **1123 BCM**
 - Surface water: **690 BCM**
 - Groundwater: **433 BCM**

Water Resource Development and Utilization

- Objective:
 - Drinking water
 - Industrial use
 - Agricultural demands
 - Environmental issues
- Factors affecting water resources:
 - **Climatic:** rainfall intensity, duration, distribution, snow, evapotranspiration
 - **Physiographic:** land use, soil type, slope, drainage, stream density
 - **Geological:** rock composition, permeability, porosity, transmissivity

Irrigation Development and Potential

- Irrigation history dates to prehistoric times
- Vedas and ancient scriptures mention wells, canals, tanks
- Out of **329 million ha** geographical area:
 - Cultivable area: **182 million ha**
 - Net sown area: **140 million ha**
 - Irrigated area: **62 million ha (~44%)**
- Irrigation potential:
 - 1951: **22.6 million ha**
 - End of Eleventh Plan: **108.91 million ha**
- Irrigation projects designed for **75% dependability**

Future Water Requirements

- Projected irrigation demand by 2050: **1180 BCM**
- Present water use:
 - Irrigation: **~83%**
- Per capita water availability:
 - 1951: **5177 m³**
 - 2010–11: **1545 m³**

National Water Policy (2012)

Salient Features

- Integrated perspective (local to national)
- Equity and social justice in water allocation
- Transparent and participatory governance
- Water as **common pool community resource**
- Priority to:
 1. Drinking water & sanitation
 2. Domestic needs
 3. Food security

4. Ecosystem needs

- River basin as basic hydrological unit
- Demand management emphasized
- Efficient water use and pollution reduction
- Climate change impacts to be incorporated
- Regulation of water-using activities

Participatory Irrigation Management (PIM)

- Involvement of farmers in:
 - Planning
 - Design
 - Construction
 - Operation & maintenance
 - Distribution
 - Financing
- Two strategies:
 1. **Legislative** – Government-led
 2. **Motivational** – Farmer-led
- Organizations: **Water Users' Associations (WUA)**
- Objective:
 - Better availability and utilization of water
 - Decentralized decision-making

Environmental Impacts of Irrigation Projects

- Changes in soil and water quality
- Waterlogging and salinization
- Groundwater pollution (fertilizers, pesticides)
- Reduced return flows
- Reduced biodiversity
- Over-extraction → water table decline, land subsidence, seawater intrusion

- Energy consumption and social impacts
- Requires mitigation at planning stage

Land-Use Pattern

- Reflects:
 - Physical land characteristics
 - Institutional and economic framework
- Helps identify:
 - Shifts in land use
 - Economic development trends

Land-Use Data (India)

- Total geographical area: **328 million ha**
- Land utilization data available for **~93%**
- Trends:
 - Increase in non-agricultural use
 - Increase in net area sown
 - Increase in cropping intensity due to irrigation expansion

Standard Land-Use Classifications

- Forests
- Land under non-agricultural uses
- Barren & unculturable land
- Permanent pastures & grazing lands
- Miscellaneous tree crops
- Culturable wasteland
- Current fallows
- Other fallows
- Net area sown

Irrigated Area Definitions

- **Net irrigated area:** irrigated once in a year

- **Total net unirrigated area:** net sown – net irrigated
- **Gross irrigated area:** irrigated more than once
- **Cropping intensity:**

$$\text{Cropping Intensity} = \frac{\text{Gross cropped area}}{\text{Net area sown}}$$

Land-Use Planning

- Based on present land-use patterns and trends
- Requires:
 - Institutional arrangements
 - Government policies
 - Agro-economic zone assessment
- Aim: efficient and sustainable land utilization

Land Capability Classification

Definition

Systematic arrangement of land based on properties determining its ability for sustained use.

- Classified into **8 classes**
- **Class I–IV:** Cultivation suitable
- **Class V–VIII:** Not suitable for cultivation

Land Capability Classes

Class I

- Nearly level
- Deep (>90 cm), well-drained soils
- Very few limitations
- Wide range of crops

Class II

- Moderate restrictions
- Gentle slope, moderate erosion

- Requires simple conservation practices

Class III

- Moderate to severe restrictions
- Steeper slopes, erosion hazards
- Requires special conservation measures

Class IV

- Severe limitations
- Very restricted agriculture
- Best for perennial vegetation

Class V

- No erosion hazard
- Cultivation not feasible
- Suitable for pasture

Class VI

- Severe erosion
- Steep slopes, shallow soils
- Grazing and forestry only

Class VII

- Very severe limitations
- Shallow, stony, gullied lands
- Not practical for cultivation

Class VIII

- Completely barren
 - Suitable only for recreation, wildlife, water supply
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